

ESPRIT · 2026

SOFTWARE ENGINEERING

PEAXIS

A Modular AI-Powered Hiring Operating System

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AI-Assisted Recruitment

Explainable AI

Multi-Tenant SaaS

Modular Architecture

1 INTRODUCTION & CONTEXT

Presentation Plan

An engineering defense: from problem framing to validated implementation.

01 Context & Scope

The hiring problem, internship scope, and the project objective.

02 Requirements & Solution

Product requirements and the PEAXIS platform modules.

03 Architecture

System boundaries, deployment topology, and design decisions.

04 AI Engineering

Parsing, evidence matching, deterministic scoring, and explainability.

05 Validation & Findings

Tests, delivery evidence, limitations, and hardening priorities.

06 Conclusion

What was implemented, what was learned, and what comes next.

2 INDUSTRY & PROBLEM

Recruitment in the Digital Era

Four forces are reshaping how organizations hire.



Application volume

Online job boards and easy-apply flows multiply applications per opening.



Recruiter workload

The same recruiter teams review a growing volume of candidates per role.



Candidate expectations

Applicants expect fast feedback, transparency, and a clear application status.



AI adoption in HR

Hiring teams increasingly expect AI support in screening and decision-making.

- Hiring has become a higher-volume, higher-expectation process — this is the environment in which recruitment software now has to operate.

2 COMPANY PRESENTATION

Prospecter — Internship Host Company

AI-powered B2B platform for outbound prospecting.

MISSION

Autonomous AI SDRs help B2B teams scale outbound prospecting.

AI / LLM

B2B SAAS

SALES AUTOMATION

OUTBOUND GTM



AI SDR Engine

Autonomous AI sales rep — qualifies leads and orchestrates outreach sequences at scale.



Lead Intelligence

Enriches prospects with firmographics and intent signals to prioritise high-value targets.



Outbound Automation

Multi-channel sequences with LLM personalisation — zero manual prospecting.



Pipeline Analytics

Real-time conversion tracking, A/B message testing, and CRM sync.



Strategic context — LLM pipelines, multi-tenancy, Redis, and background workers informed PEAXIS.

3 INTERNSHIP

Internship Context

Software Engineering internship at **Prospecter**, an AI-powered B2B platform.



Full-Stack Engineering

Production Next.js and NestJS codebases, with tenant-scoped data models.



AI Integration

LLM pipelines for parsing, classification, and structured generation.



Platform Engineering

Docker deployments, async queues, and semantic search infrastructure.

Scope: PEAXIS is an independent product, built to the production standards observed during the internship.

SOFTWARE ENGINEER

AI INTEGRATION

SOLO CONTRIBUTOR

2 INDUSTRY & PROBLEM

Current recruitment challenges

The scale of the hiring problem, before the engineering response.

42 days

Average time-to-hire
SHRM 2024

75%

Recruiters overloaded
LinkedIn Talent Report

60%

Candidates report no feedback
Indeed Survey 2023



Manual screening

Reviewing every CV by hand does not scale with rising application volume.



Keyword-based ATS

Legacy filters match exact keywords and miss well-qualified, non-standard CVs.



Inconsistent evaluation

Criteria vary by reviewer, with limited traceability of how a decision was reached.



Candidate experience

Long application forms and no status feedback push qualified talent away.

Cost of a bad hire: averages **\$14.9K**; strong candidates can accept competing offers within days.

Engineering response: keep screening responsive, make every score reviewable, and keep candidates informed.

4 EXISTING SOLUTIONS & GAP

The gap: **integrated, explainable AI**



AI black boxes

Recommendations need traceability: requirements, cited evidence, and recruiter review.



Fragmented tooling

Specs, pipelines, calendars, and notes live in separate tools.



No candidate intelligence

Tools track applications but rarely guide decisions or interviews.

Evidence first

Every assessment links a requirement to candidate evidence.

Human authority

AI assists evaluation; recruiters decide the hiring outcome.

Durable operations

Slow AI work runs outside the request path with explicit states.



This gap defines the project's objective

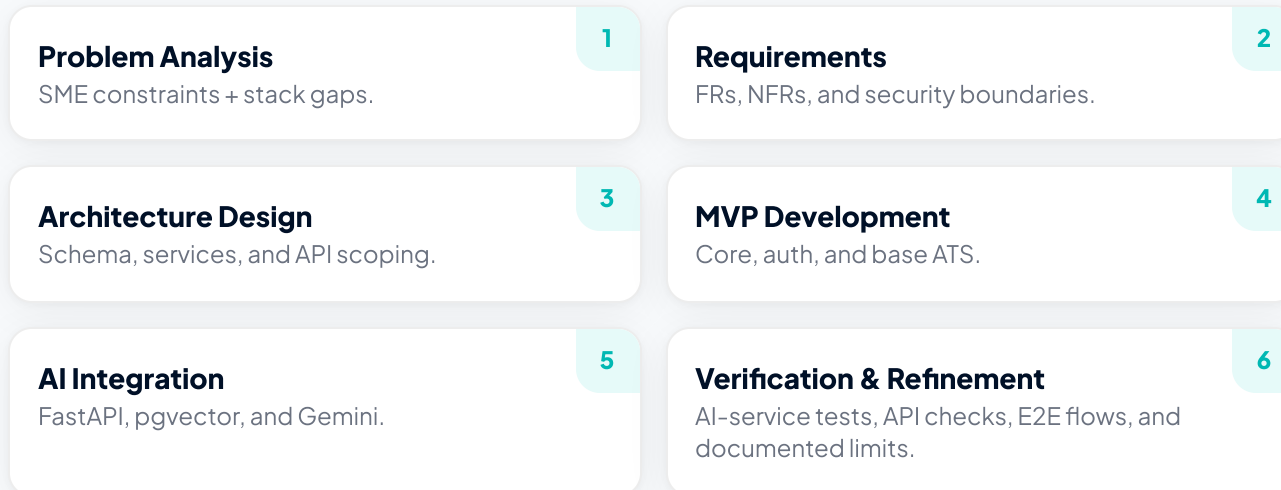
Integrated workflows, evidence-based scoring, and a candidate-facing experience in one auditable system.

5 METHODOLOGY & REQUIREMENTS

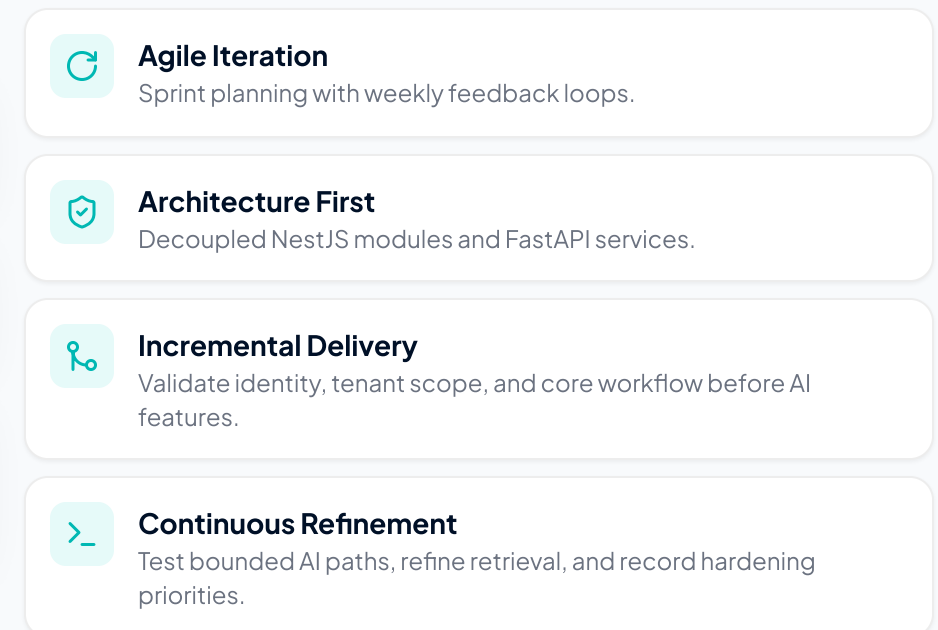
Engineering Methodology

From problem framing to a tested implementation.

PROJECT ENGINEERING PIPELINE



EXECUTION STRATEGIES



5 METHODOLOGY & REQUIREMENTS

Core platform requirements

Five delivered foundations for the recruitment workflow.



FR-01

Authentication & Security

- Argon2 password protection
- JWT + OAuth sign-in



FR-02

Multi-Tenant Architecture

- Business-scoped records
- Tenant context enforced



FR-03

Roles & Permissions

- Role and plan guards
- Feature access by subscription



FR-04

Job Management

- Create and filter jobs
- Hybrid text + vector search



FR-05

Application Pipeline

- Tracked application stages
- Async assessment updates

5 METHODOLOGY & REQUIREMENTS

AI workflow requirements

Implemented AI capabilities — no scheduling or dashboard claims.



FR-06

Evidence Matching

- Requirement-level assessment
- 0-100 alignment score



FR-07

CV Parsing

- PDF / DOCX / TXT upload
- Async structured extraction



FR-08

Candidate Onboarding

- Apply with a CV
- Confirm parsed profile



FR-09

Explainable Review

- Cited strengths and gaps
- Verification state per requirement



FR-10

Reliable AI Processing

- Queued AI work items
- Retry and failure status

5 METHODOLOGY & REQUIREMENTS

Implemented **quality attributes**

Delivered engineering qualities; performance SLAs remain to validate.



Container-ready

NFR-01 **Deployment Readiness**

Web, API, worker, AI runtime, and data services are separated.



Implemented controls

NFR-02 **Security**

Auth, file gates, rate limits, tenant scoping, and service secrets.



No AI in request path

NFR-03 **Responsive Requests**

Parsing and assessment run outside the HTTP request path.



Explicit work states

NFR-04 **Failure Handling**

Retries and durable work states expose processing failures.



Cited assessment trail

NFR-05 **Explainability**

Scores retain evidence, gaps, and verification state.



Modular platform

NFR-06 **Modularity**

Separate web apps, API, worker, and inference boundary.

6 PROPOSED SOLUTION

PEAXIS — Modular AI Hiring Operating System

Three modules, one platform data layer, and a bounded AI service.



Operational Backbone

PEAXIS Core

Identity, tenant context, roles, and plan controls.



Recruiter Execution

PEAXIS Hire

Candidate pipeline, profile review, and evidence assessment.



Talent Acquisition

PEAXIS Jobs

Job discovery, CV parsing, and candidate applications.

BUSINESS ADMIN

PEAXIS Core

Access, tenant context, plans

RECRUITER / HR TEAM

PEAXIS Hire

Pipeline and evidence review

CANDIDATE / TALENT

PEAXIS Jobs

Search, profile, application

- The platform API owns transactional data through Prisma → PostgreSQL

NEXT.JS 16

NESTJS

FASTAPI

POSTGRESQL + PGVECTOR

REDIS

BULLMQ

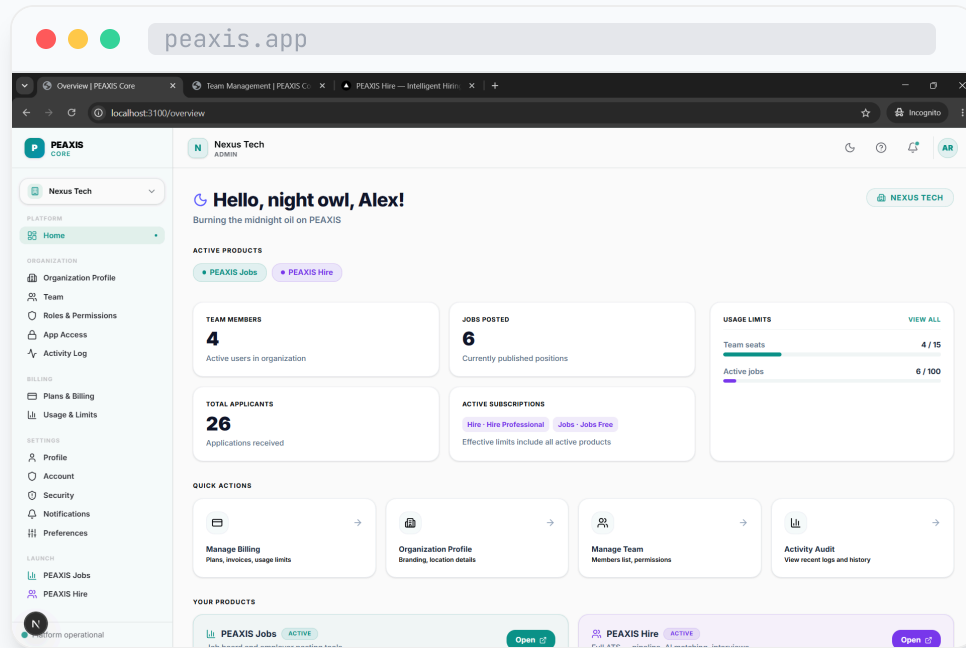
GEMINI / AZURE OPENAI

STRIPE

6 PROPOSED SOLUTION

PEAXIS Core

The shared business foundation every module relies on.



PEAXIS Core — access, business context, and plan controls



Identity & Access

Authentication, business scoping, and role-based permissions.



Plans & Entitlements

Subscription plan controls and feature access.



Business Context

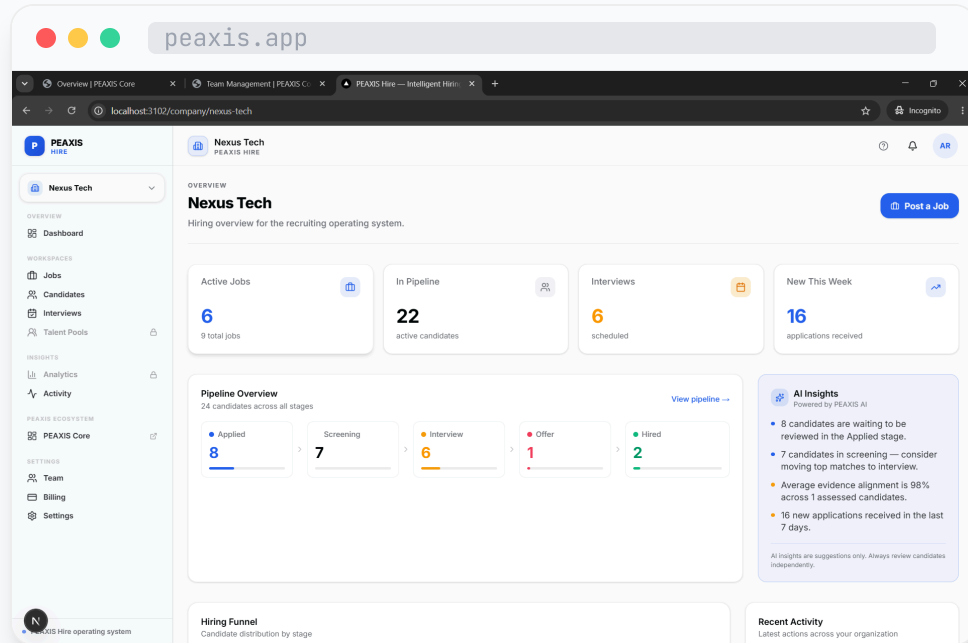
Business settings and member-level access.

One control center for tenancy, access, and billing across every module.

6 PROPOSED SOLUTION

PEAXIS Hire

The recruiter workspace for pipeline review and explainable assessment.



PEAXIS Hire — recruiter pipeline and assessment workspace



Pipeline workspace

Track applications and open each candidate profile.



Evidence assessment

Review score, cited evidence, gaps, and verification state.



Async processing

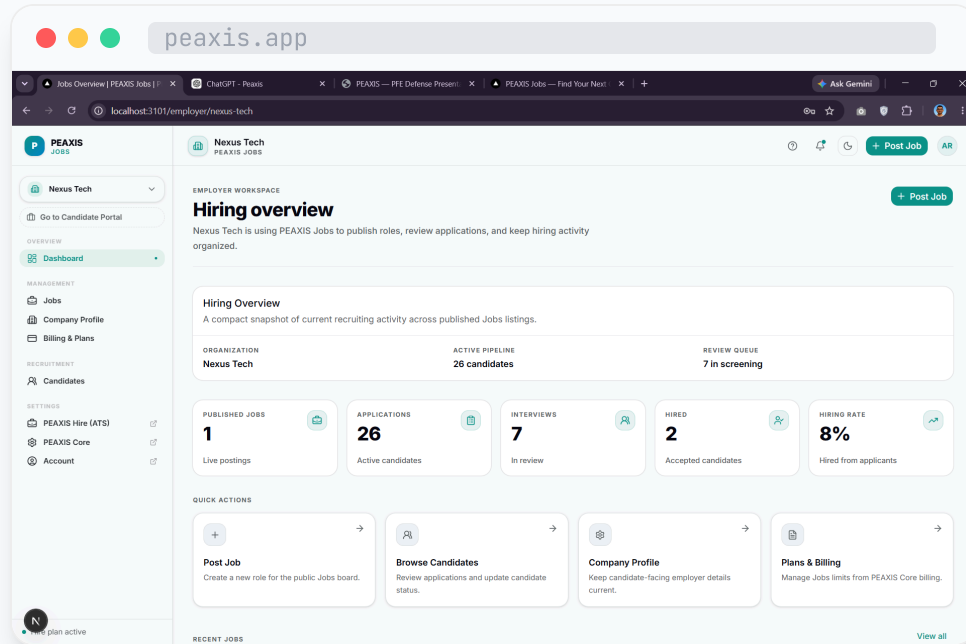
AI work progresses with visible pending, retry, and failure states.

One place to inspect the pipeline and justify an assessment with evidence.

6 PROPOSED SOLUTION

PEAXIS Jobs

The candidate portal for discovery, profile completion, and applications.



PEAXIS Jobs — job discovery and candidate application portal



Job Discovery

Hybrid full-text and vector search find relevant jobs.



Application UX

CV parsing helps construct a candidate profile.



Application Progress

Application status is visible after submission.



A clear, structured path from job discovery to application submission.

6 PROPOSED SOLUTION

AI ownership boundaries

AI infers; the platform owns rules and records; recruiters decide.



NestJS API

Owns authorization, tenant scope, business records, deterministic assessment rules, and persistence.



Dedicated worker

Executes durable AI work, applies retry rules, and reconciles queued work without blocking HTTP requests.



FastAPI service

Performs parsing, embeddings, citation-bound classification, and supported content generation only.



PostgreSQL + pgvector is authoritative for records, snapshots, and vectors.



Redis + BullMQ carries queues, locks, heartbeats, and non-authoritative caches.

NESTJS DECIDES BY RULES

FASTAPI INFERS

RECRUITER DECIDES TO HIRE

7 ARCHITECTURE & TECHNOLOGIES

Tech Stack — Four Layers

Implemented application, platform, inference, and data boundaries.

Frontend

Specialized Core, Hire, Jobs, Admin, and Landing applications share React/TypeScript patterns

Next.js 16

React

Tailwind

TypeScript

API

NestJS + Prisma: domain modules, policy enforcement, and transactional platform data

NestJS

Prisma

JWT

TypeScript

AI Service

Stateless inference for parsing, classification, embeddings, and structured generation

FastAPI

Gemini / Azure

Python

Redis

Database

Authoritative data + pgvector retrieval, Redis, and one durable BullMQ queue

PostgreSQL

pgvector

Redis

BullMQ

TENANT CONTROLS

Business-scoped domain records with explicit remediation items

ASYNC PROCESSING

BullMQ + background workers

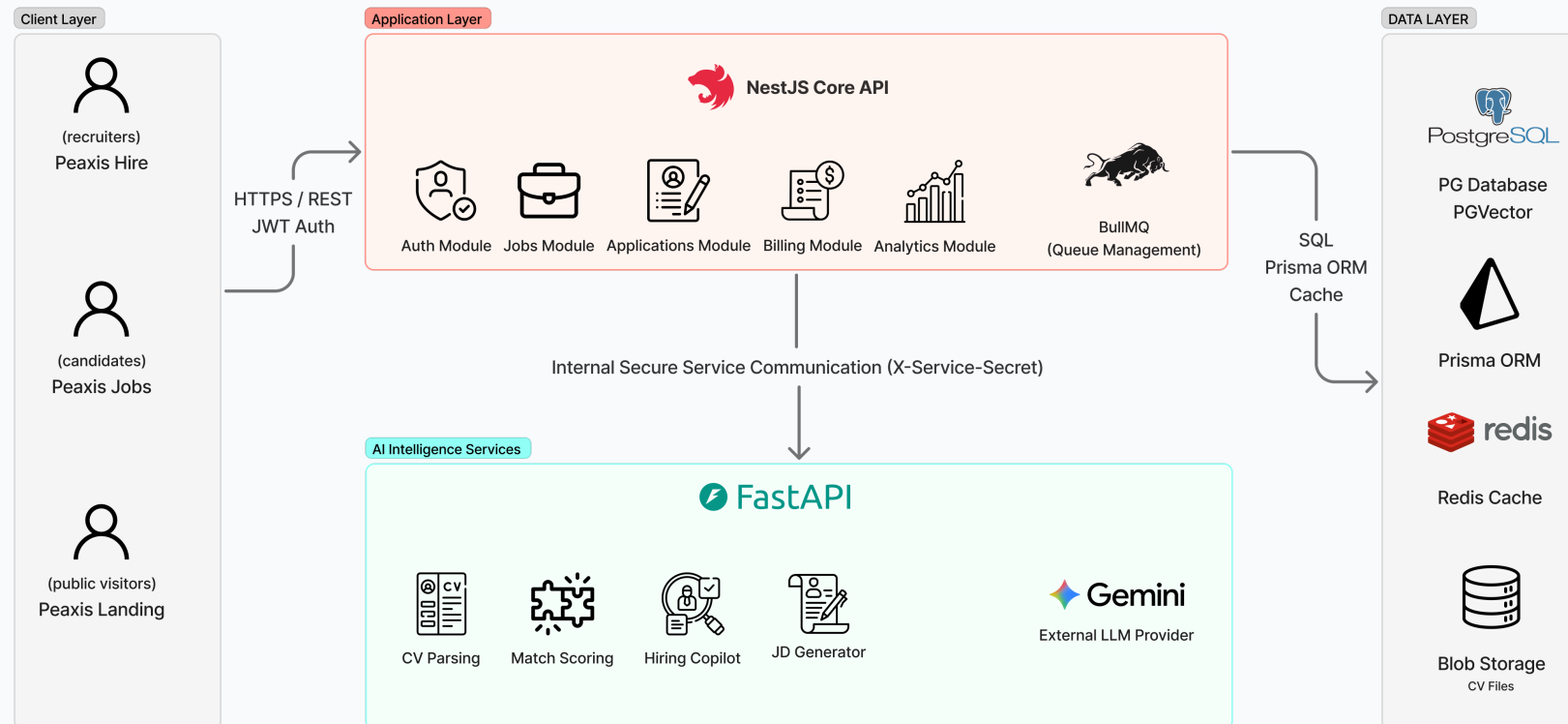
RETRIEVAL

pgvector supports job discovery and cited evidence

7 ARCHITECTURE & TECHNOLOGIES

Logical Architecture

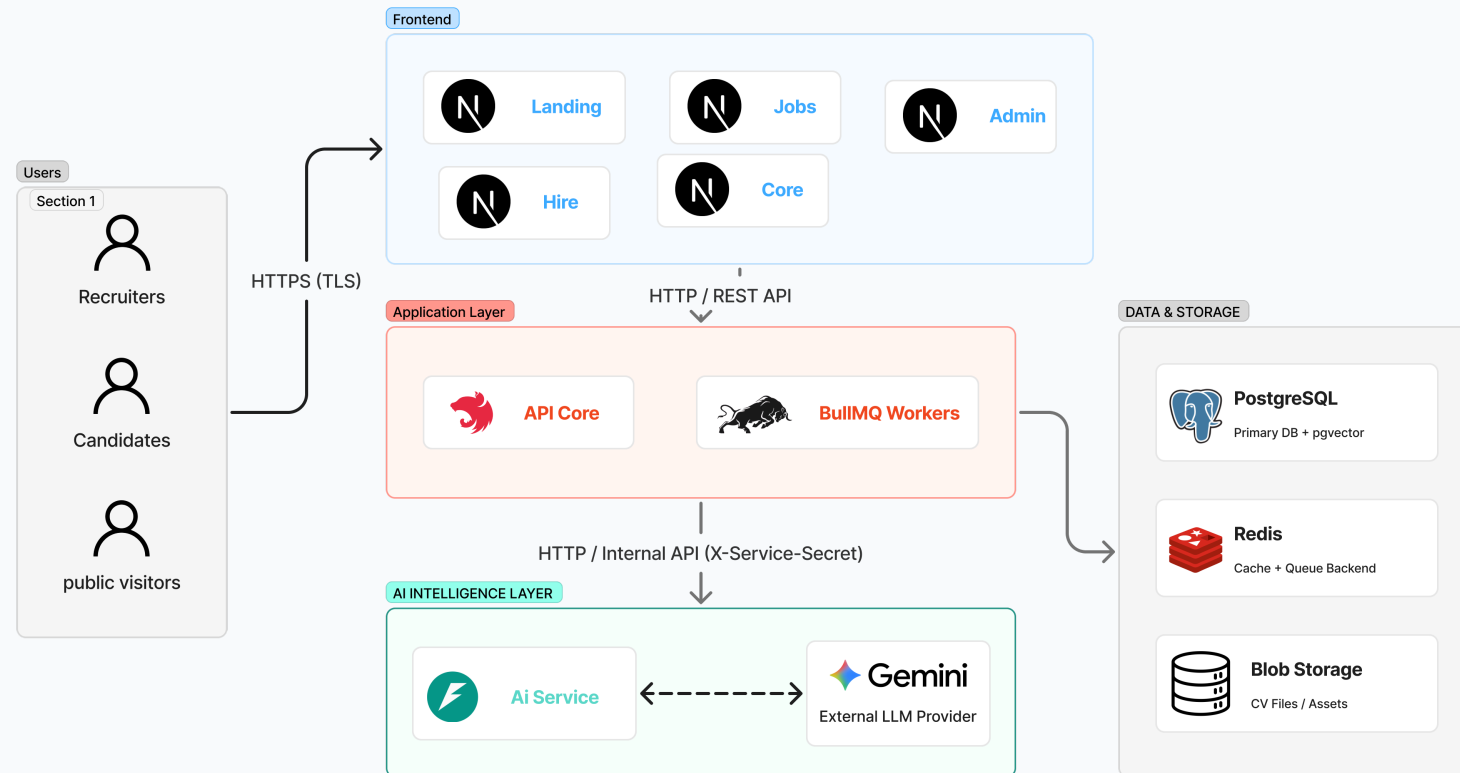
Client, platform, AI, and data boundaries. Gemini or Azure OpenAI is configured at startup.



7 ARCHITECTURE & TECHNOLOGIES

Physical Architecture

Deployment topology for web apps, platform services, AI inference, and storage. Gemini or Azure OpenAI is configured at startup.



8 ENGINEERING DEEP DIVE

Architecture Decisions

Selected for clear responsibilities and delivery control.

Modular monolith

vs microservices

Domain stays cohesive; less distributed complexity.

Separate AI service

FastAPI

Python ML stack isolated from core API load.

Specialized web apps

Next.js

Separate experiences for platform, recruiter, candidate, administration, and landing.

PostgreSQL

Prisma

Relational workflows with strong consistency.

BullMQ

async

Heavy AI work moves outside request lifecycle.

pgvector

retrieval

Retrieval for job discovery and cited evidence.

8 ENGINEERING DEEP DIVE

AI Runtime Architecture

NestJS orchestrates, FastAPI infers, and PostgreSQL is the source of truth.



Synchronous request path

Validate, authorize, persist business state, and return a processing response.
Short generation or extraction helpers can call FastAPI directly.

Asynchronous worker path

Call FastAPI, persist parse/embedding/assessment outputs, and update the durable work and application state.

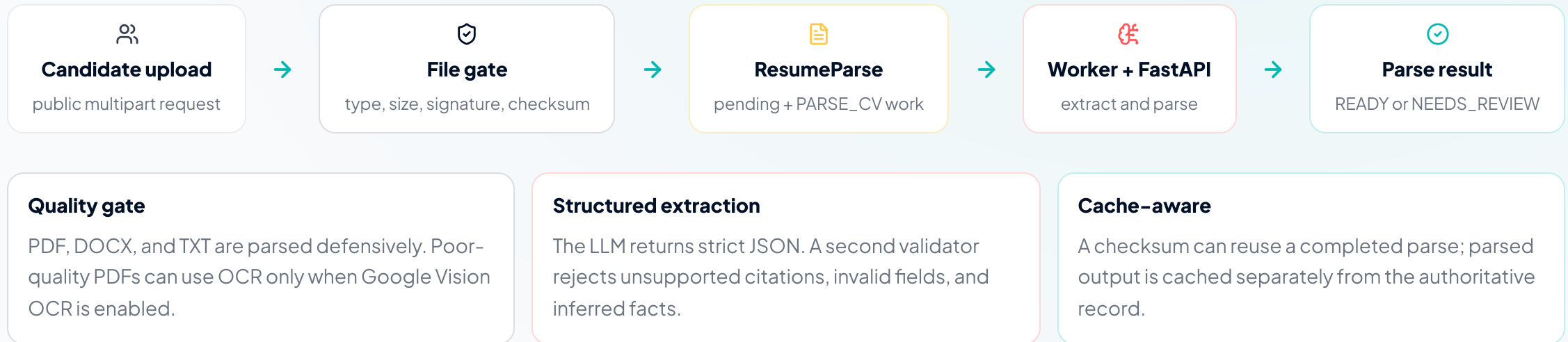
Return path: the worker writes results to PostgreSQL; Redis is used for queues, locks, heartbeats, and caches—not authoritative AI state.

8 ENGINEERING DEEP DIVE

CV Parsing Pipeline

The CV is validated and parsed asynchronously into candidate evidence.

1. INTAKE AND DURABLE PARSING



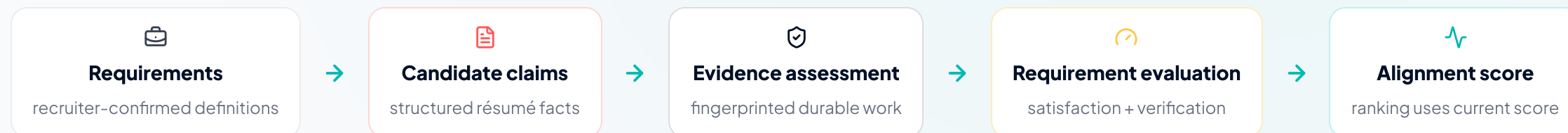
2. CANDIDATE CONFIRMATION AND RETRIEVAL PREPARATION



8 ENGINEERING DEEP DIVE

Evidence-Based Matching Engine

The model classifies cited evidence; the backend makes the reviewable decision.



AI responsibility

When direct evidence is absent, retrieve up to eight similar candidate chunks, then classify only the supplied citation set.

Backend responsibility

Exact matching, experience-month calculation, score weights, satisfaction, and verification states are deterministic NestJS rules.

Human responsibility

Recruiters define requirements, inspect evidence, and decide the candidate outcome. No AI hiring decision exists.

The current alignment and ranking score are the same persisted calculation—not a separate opaque candidate similarity model.

How Matching Is Computed

A deterministic evidence algorithm; vectors and LLMs only retrieve or classify citations.

1. Find candidate evidence

Canonical token and simple stem matching compare each requirement with source-cited claims. The fallback searches displayable résumé summary, headline, experience, and project text. Experience requirements merge overlapping date intervals into relevant months.

2. Use vectors only when needed

For unresolved COMPETENCY, RESPONSIBILITY, or TOOL_SYSTEM requirements, FastAPI embeds the requirement. pgvector retrieves only this candidate's chunks at distance ≤ 0.55 (maximum 8). The classifier sees at most 12 claims plus retrieved chunks and may cite only one supplied source.

3. Apply deterministic requirement states

No evidence, mandatory
REQUIRES_VERIFICATION

Evidence but years unmet
PARTIALLY_SATISFIED

No evidence, preferred
UNKNOWN

Evidence + policy satisfied
SATISFIED

4. Persist explainable results

An **EvidenceAssessment** stores input snapshots and the score. Each **RequirementEvaluation** retains satisfaction, constraints, and positive cited evidence.

CURRENT GENERAL: V1 SCORE

$$\text{score} = \text{round} \left(\frac{\sum (\text{weight} \times \text{state value} \times \text{evidence credit})}{\sum (\text{weight})} \right)$$

State value: SATISFIED = 100, PARTIALLY_SATISFIED = 60, all other states = 0. The current ranking score equals this alignment score; no separate opaque cosine ranking is used.

required 0.55

experience 0.25

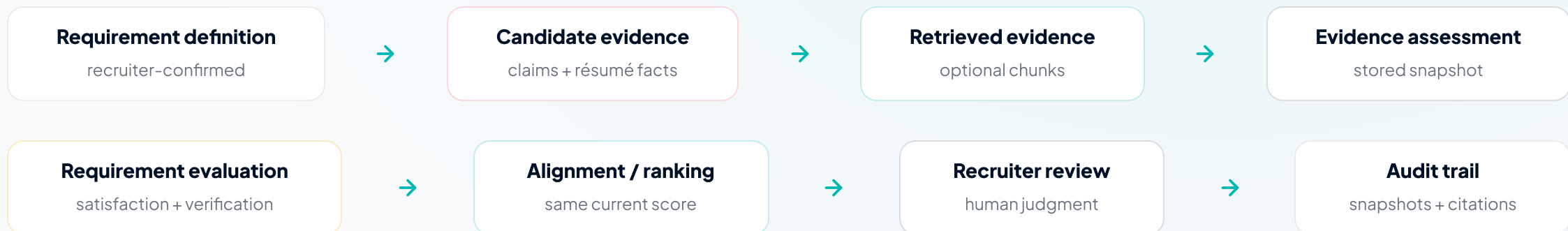
preferred 0.10

TRANSFERABLE $\times 0.60$

8 ENGINEERING DEEP DIVE

Explainable AI

Each score traces a requirement to cited candidate evidence.



The LLM can classify supplied citations; NestJS applies scoring and verification rules. Recruiter review is stored, while score-changing override policy remains future work.

Implementation Status

Delivered capabilities are separated from production-hardening gaps.



CV evidence pipeline

Implemented

Validated uploads, durable parsing, candidate confirmation, cited claims, and persisted evidence chunks.



Evidence assessment

Implemented

Requirement-by-requirement evaluation, deterministic alignment score, and persisted reviewable output.



Durable AI runtime

Implemented

AiWorkItem state, queue delivery, worker retries, and reconciliation keep inference off the request path.



Recommendation scope

Hardening

Candidate recommendation visibility needs stricter business and application constraints.



Review policy

Partial

Reviewer decisions are stored, but override effects are not yet part of the versioned scoring policy.



Data governance

Hardening

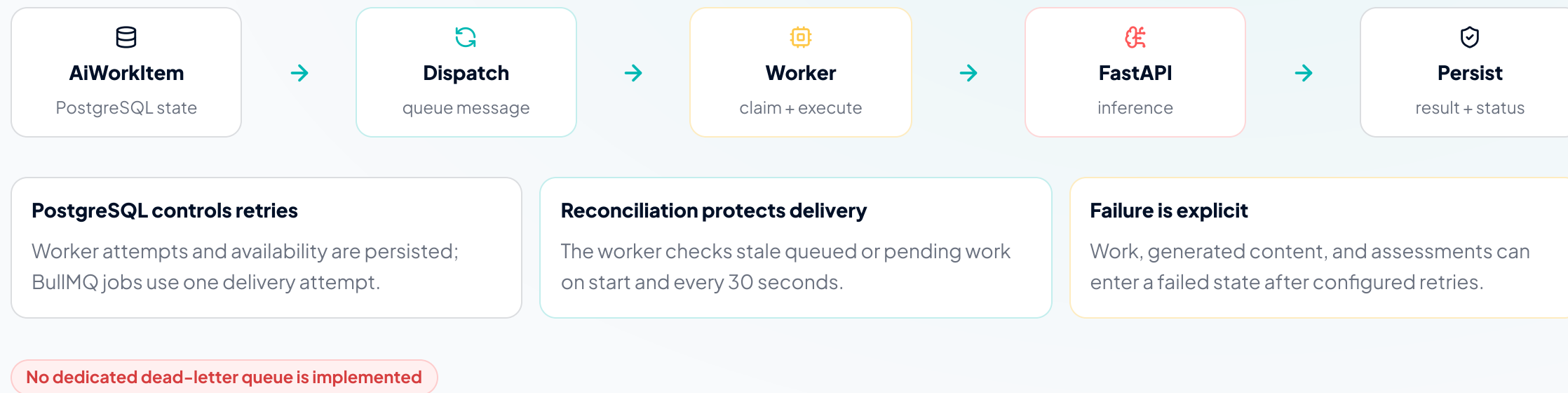
Provider retention, consent, PII redaction, and stale-vector lifecycle policy are still to be formalized.

This distinction keeps the technical defense accurate: a working capability is not presented as a completed production assurance.

8 ENGINEERING DEEP DIVE

Background Processing

PostgreSQL owns work state; BullMQ sends execution to the worker.



8 ENGINEERING DEEP DIVE

Security Architecture

Identity, tenant, file, and service controls — with known gaps explicit.



JWT

Authenticated API access



Membership

Active tenant membership in protected flows



businessId

Tenant-scoped domain records



Rate limits

Public CV and FastAPI request throttling



File gates

Size, MIME, extension, and signature checks



Service secret

FastAPI internal-call authentication



Provider PII

Raw CV content is sent for inference



Visibility gap

Candidate recommendation access needs remediation

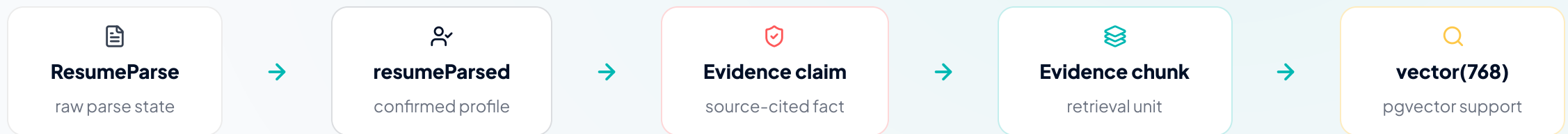
Pre-production hardening: enforce active membership for generated content and define provider retention, consent, and PII redaction policy.

8 ENGINEERING DEEP DIVE

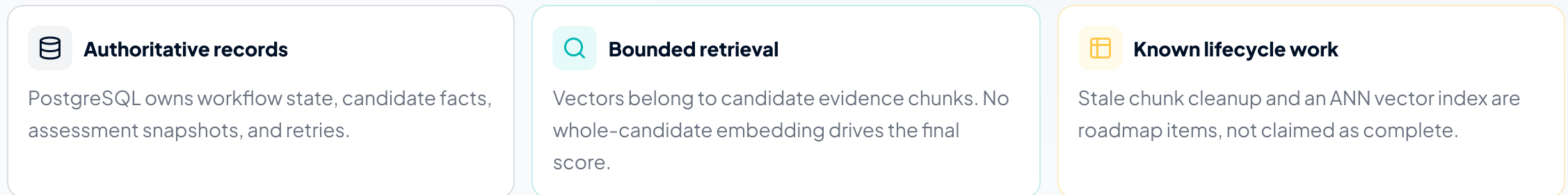
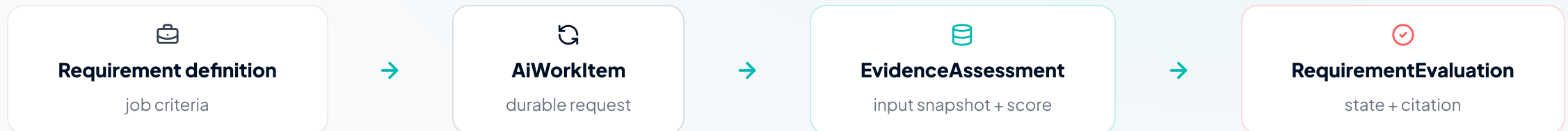
AI Data Model

PostgreSQL stores the evidence trail; pgvector supports retrieval, not decisions.

CANDIDATE EVIDENCE PATH



ASSESSMENT PATH



Verification Evidence

Unit, service, E2E, and delivery checks; unmeasured claims stay unclaimed.

Verification layer	Latest evidence	Scope
FastAPI service	18 Python tests passed locally	CV parsing, provider routing, sanitization, and website-profile helpers
Platform API	Unit, API E2E, and Playwright workflows are configured	Platform behavior and critical recruiting journeys
Delivery pipeline	GitHub Actions launch-readiness workflow	Install, typecheck, build, tests, API E2E, and browser E2E



Verified behavior

Provider selection, parsing safeguards, and bounded service behavior have executable tests.



Coverage to strengthen

Add direct tests for EvidenceMatchingService and the AI worker, including thresholds and retry transitions.



Before defense

Repair the stale CandidatesPublic unit-test setup after the RecommendationService dependency was added.

No model-accuracy, latency, throughput, or SLA value is claimed until it is measured on a controlled evaluation sample.

AI Models & Routing

One provider is selected at startup: Gemini or Azure OpenAI.

Feature	Model route	Reason
CV parsing	Configured provider	Structured résumé extraction
Evidence classification	Configured provider	Only supplied citations are classified
Embeddings	Configured embedding model	Job search and candidate evidence retrieval
Generated content	Configured provider	Drafted, reviewable content
Assessment score	NestJS + PostgreSQL	Deterministic rule application

Provider selected at startup

FastAPI rate limit: 60/min per IP

Structured JSON where supported

Raw résumé content reaches the provider

8 ENGINEERING DEEP DIVE

Performance Optimizations

Bounded retrieval, cache reuse, and durable work reduce synchronous AI latency.



Redis cache

Bounded parse, embedding, and selected-generation reuse; never the source of truth.



Durable worker

Slow inference happens outside HTTP requests with task-specific retry policies.



Hybrid job search

PostgreSQL full-text search merges with pgvector semantic results.



Chunk retrieval

Evidence retrieval is constrained to the candidate and returns only a small cited set.



Provider retries

The inference client retries transient provider/network failures.



Graceful search

If query embedding fails, public job search returns keyword results with degraded status.

Known limit: pgvector is enabled, but the baseline migration does not create an ANN vector index.

Technical Challenges & Solutions

Safeguards and limitations remain explicit for technical review.

Challenge	Solution	Outcome
AI latency	Durable work + worker retries	Responsive request path
CV reliability	File gates + quality gate + optional OCR	READY or NEEDS_REVIEW state
Grounding	Citation-bound evidence classification	No free-form evidence invention
Explainability	Snapshots, evaluations, and citations	Auditable assessment trail
Provider outage	Retries and explicit failure states	Recoverable, observable processing
Known access gap	Candidate recommendation visibility needs remediation	Documented before production

Roadmap & Future Work

Next: harden the delivered system before widening AI capability.

NEAR-TERM

Access control

Enforce active membership for all AI-generated content and correct recommendation visibility.

Atomic delivery

Commit application and assessment work atomically with a transactional outbox pattern.

Reviewer policy

Make overrides affect an explicit versioned assessment policy and explainability.

MID-TERM

FinOps coverage

Route every chargeable AI call through operations and credit ledger reconciliation.

Prompt controls

Centralize prompt construction and test every user-controlled input surface.

Vector lifecycle

Add stale-chunk cleanup and measure before introducing an ANN vector index.

LONG-TERM VISION

Governed expansion

Expose only supported FastAPI helpers through tenant-authorized NestJS contracts.

Data governance

Define provider retention, PII classification, consent, and operational redaction.

Results & Conclusion



Engineering Achievements

- A full-stack multi-tenant SaaS: Core workspace, Hire ATS, and Jobs portal
- Evidence-based assessment: cited requirements, deterministic evaluations, and an audit trail
- Durable AI processing: queued work outside the request path via a dedicated worker



Architectural Achievements

- Clear separation of concerns across frontend, API, worker, and inference layers
- Business-scoped platform controls with explicit access-control hardening priorities
- Semantic retrieval with pgvector and a containerized inference boundary



Lessons Learned

- Tenant boundaries require continuous authorization review, not only schema design
- A stateless inference boundary keeps business state and authority in one place
- Bounded AI — cited evidence, explicit failures, and human review — keeps risk manageable



PEAXIS demonstrates that AI-assisted recruitment can remain reviewable and operationally bounded — with people retaining authority over hiring decisions and known hardening work kept explicit.

Thank you Questions?

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